



Lecture 1.1: Introduction to Power System

EEE 321: Power System I

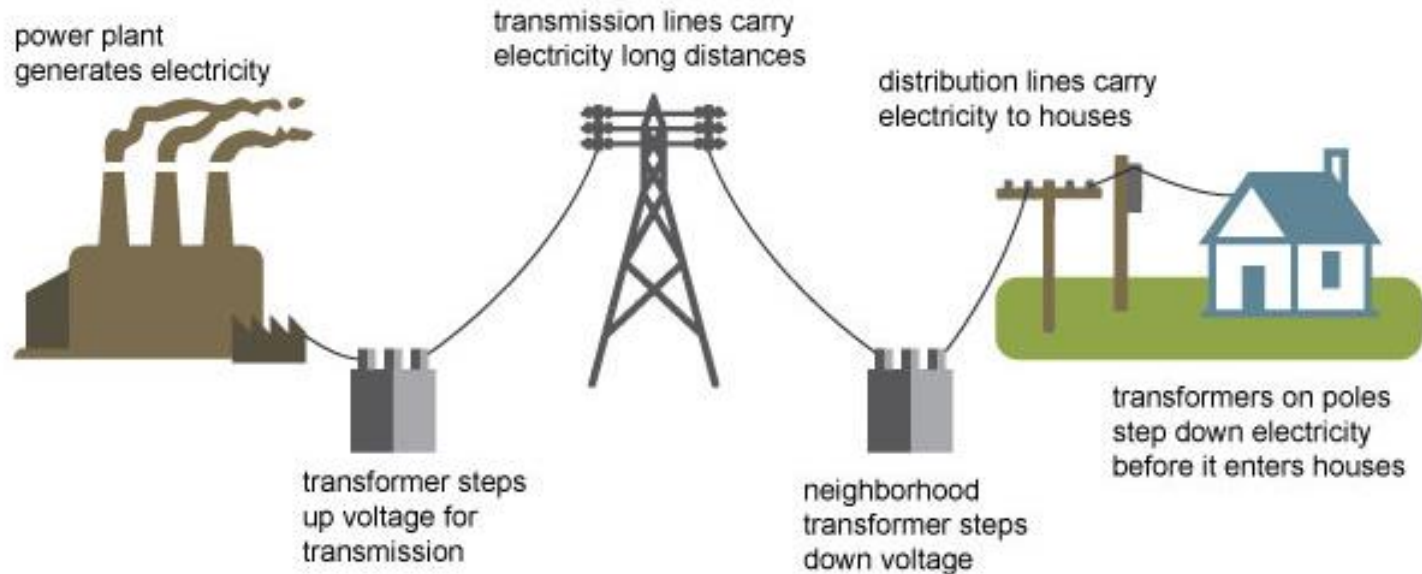
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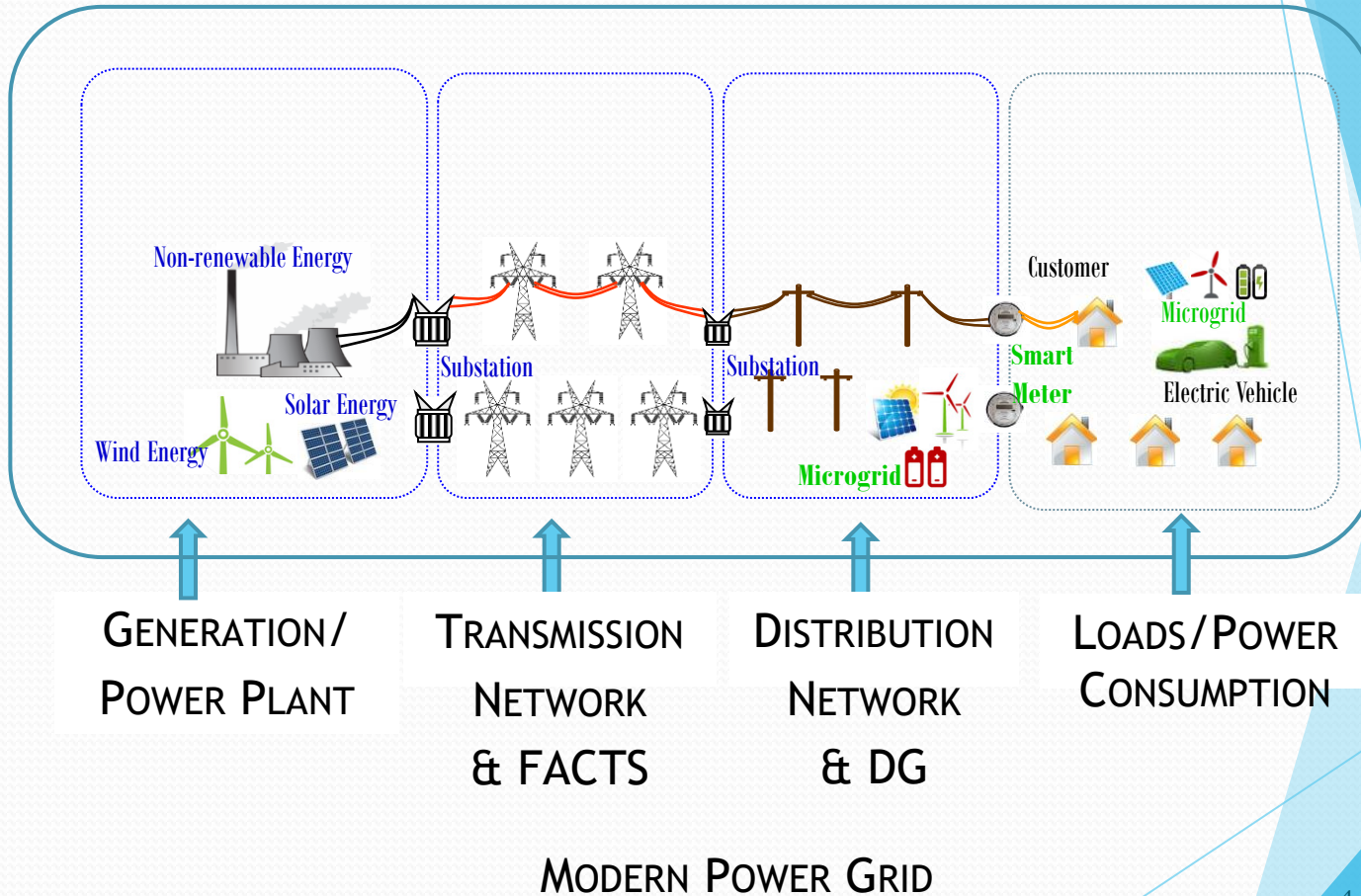
Elements of Power System

Electricity generation, transmission, and distribution



CONVENTIONAL POWER GRID

Elements of Power System



Elements of Power System

Loads

Total load demand of an area depends upon its population and the living standards of people. General nature of load is characterized by the load factor, demand factor, diversity factor, power factor and utilization factor. In general, the types of load can be divided into the following categories: (1) Domestic (2) Commercial (3) Industrial (4) Agriculture.

Domestic Load: Domestic load mainly consists of lights, fans, refrigerators, airconditioners, mixer, grinders, heaters, ovens, small pumping motors etc.

Commercial Load: Commercial load mainly consists of lighting for shops, offices, advertisements etc., fans, heating, airconditioning and many other electrical appliances used in commercial establishments such as market places, restaurants etc.

Industrial Loads: Industrial loads consists of small-scale industries, medium-scale industries, large-scale industries, heavy industries and cottage industries.

Agriculture Loads: This type of load is mainly motor pump-sets load for irrigation purposes. Load factor for this load is very small, e.g., 0.15–0.20.

Power Factor of various Loads

Induction motors	:	0.6–0.85
Fractional HP motors	:	0.5–0.80
Fluorescent lamps	:	0.55–0.90
Neon signs	:	0.40–0.50
Fans	:	0.55–0.85
Induction furnaces	:	0.70–0.85
Arc welders	:	0.35–0.55

Bangladesh Power System

Present Structure of Power Sector

Apex Institution

Power Division, Ministry of Energy & Mineral Resources (EMR)

Regulator

Bangladesh Energy Regulatory Commission (BERC)

Generation

Bangladesh Power Development Board (BPDB)

Ashuganj Power Station Company Ltd. (APSCL)

Electricity Generation Company of Bangladesh (EGCB)

North West Power Generation Company Ltd. (NWPGCL)

Independent Power Producers (IPPs)

Transmission

Power Grid Company of Bangladesh Ltd (PGCB)

Distribution

Bangladesh Power Development Board (BPDB)

Dhaka Power Distribution Company (DPDC)

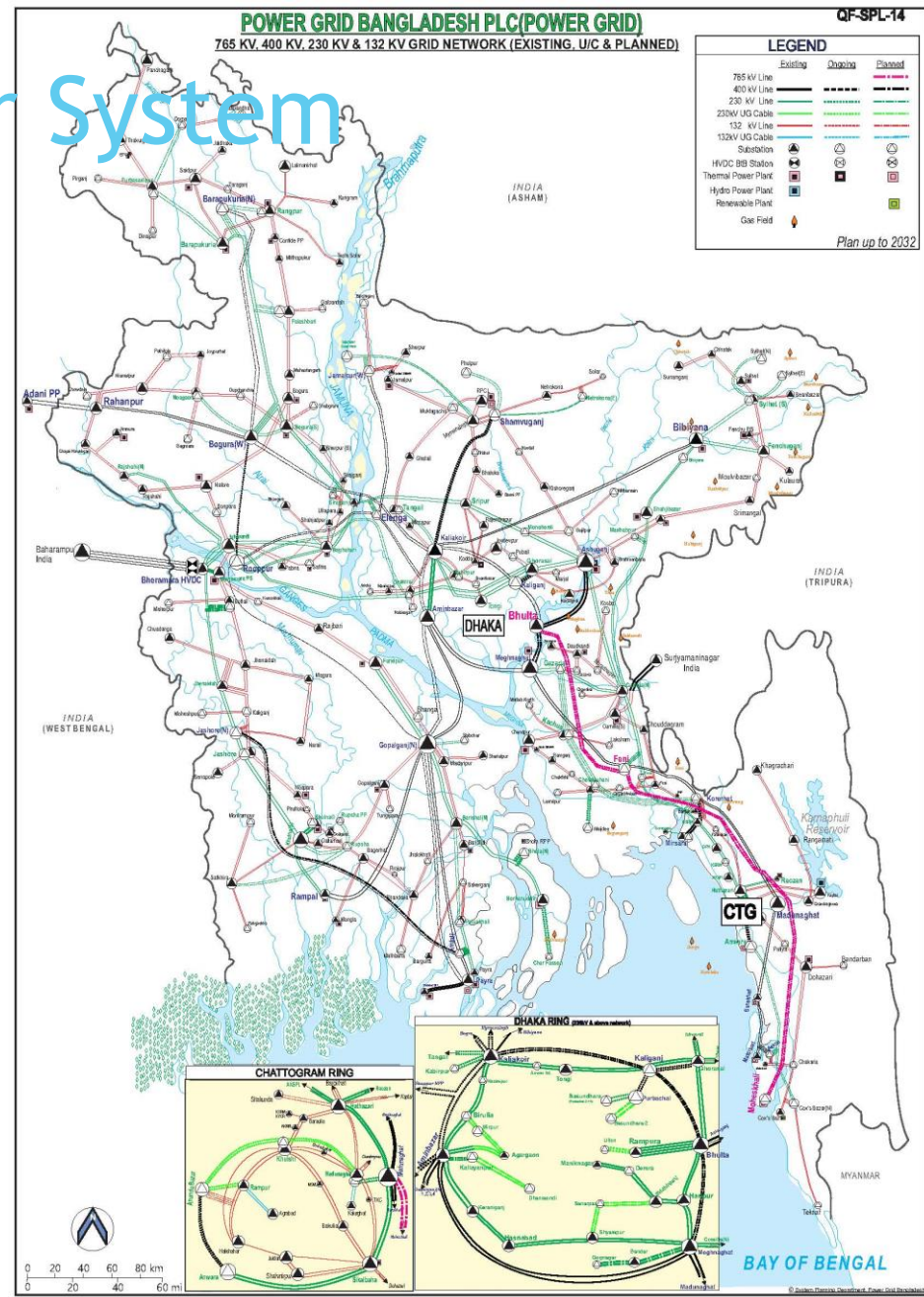
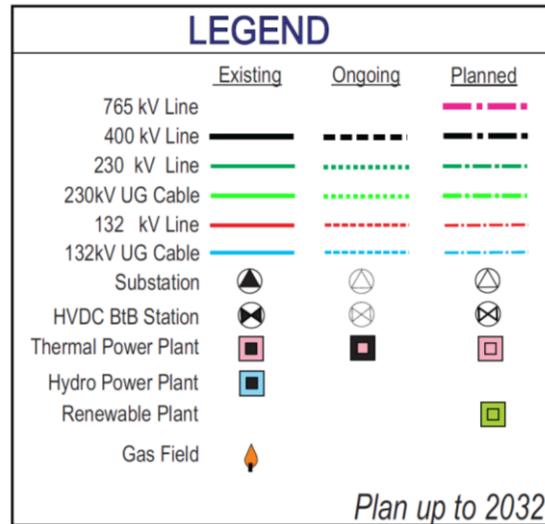
Dhaka Electric Supply Company Ltd (DESCO)

West Zone Power Distribution Company (WZPDC)

Rural Electrification Board (REB) through Rural Co-operatives

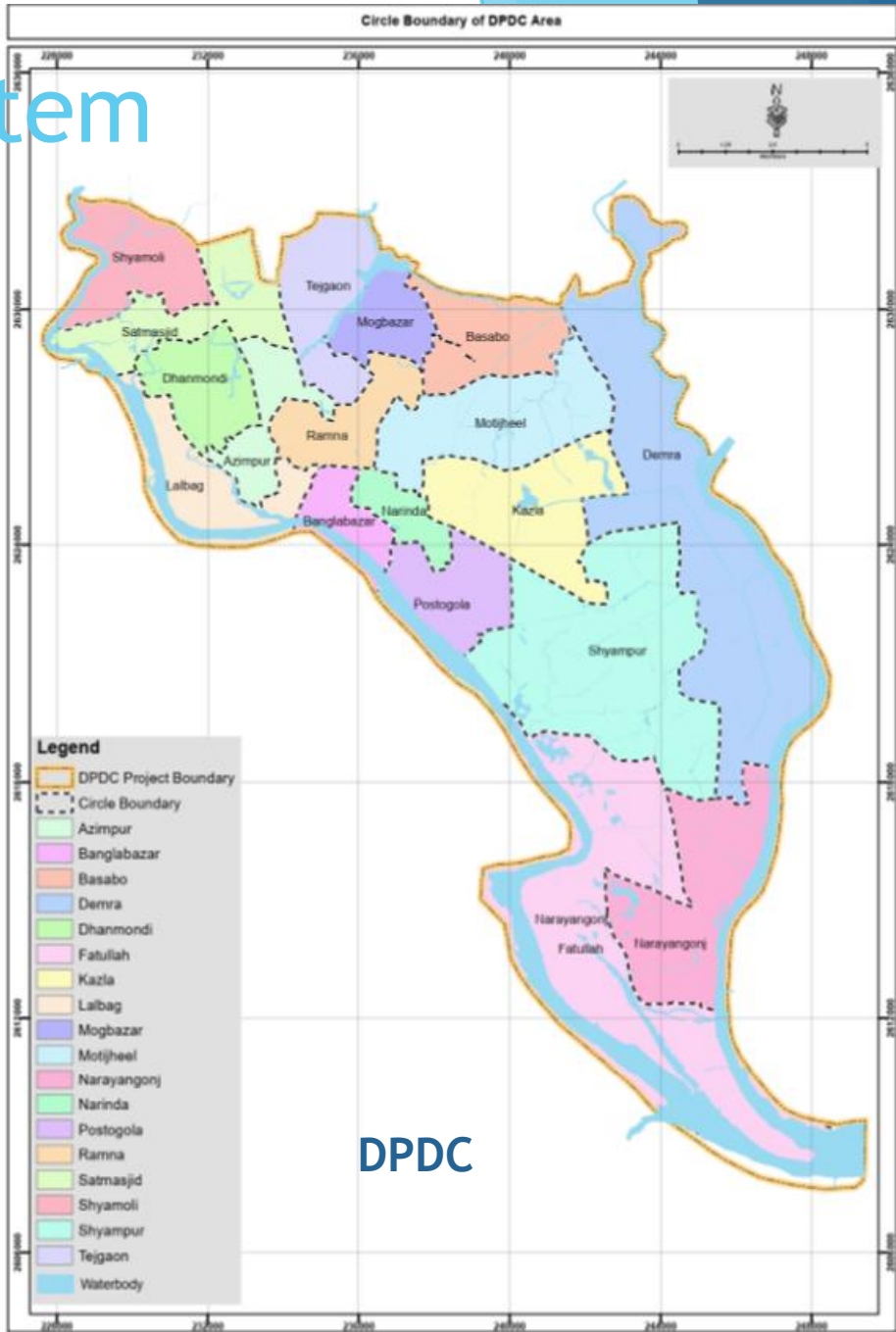
Bangladesh Power System

Power Grid of Bangladesh



Bangladesh Power System

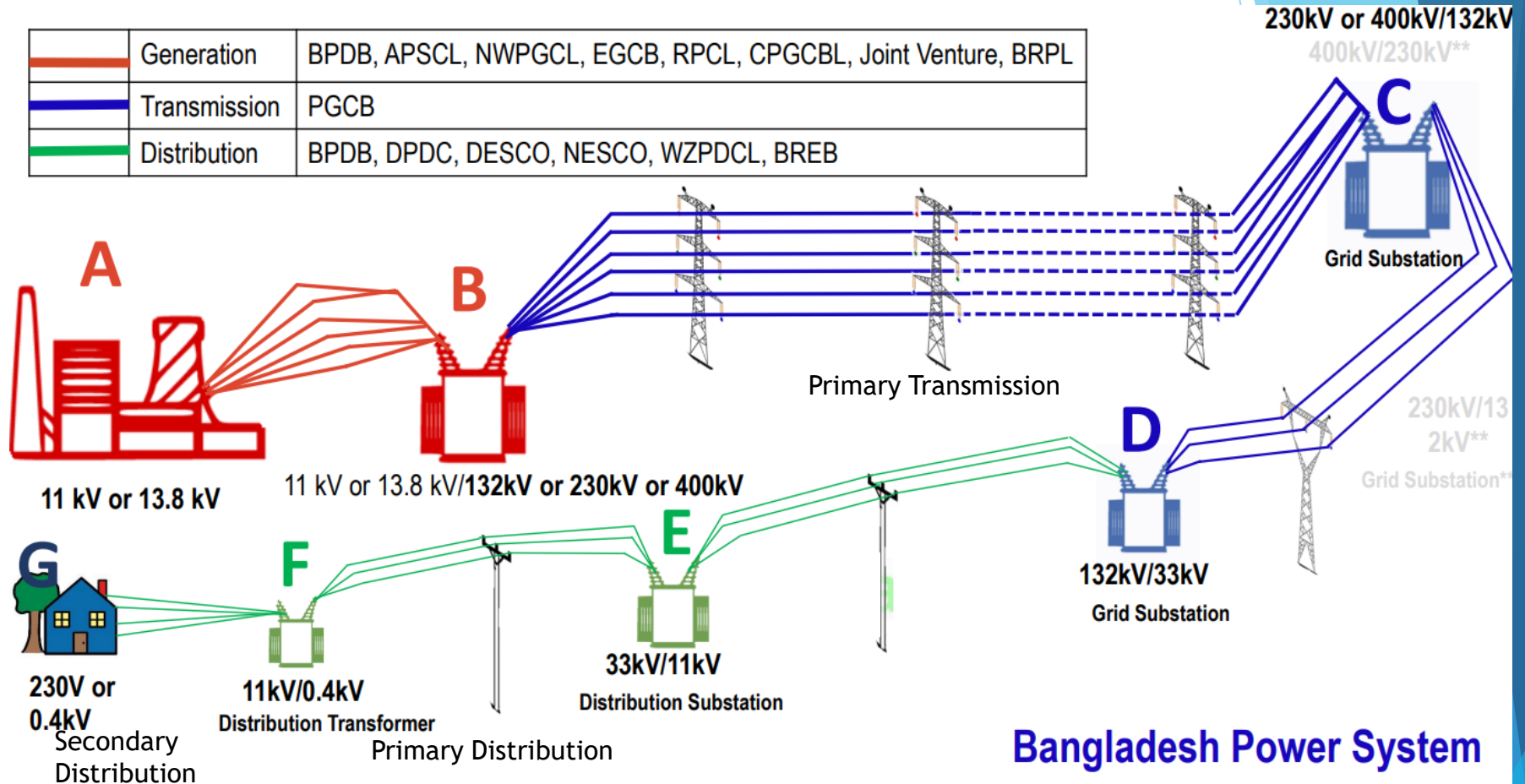
Distribution Network Area Map of Dhaka



Source: DESCO Annual Report, 2024
DPDC Annual Report, 2024.

Bangladesh Power System

	Generation	BPDB, APSCL, NWPGL, EGCB, RPCL, CPGCBL, Joint Venture, BRPL
	Transmission	PGCB
	Distribution	BPDB, DPDC, DESCO, NESCO, WZPDCL, BREB



Bangladesh Power System

Reasons behind stepping up the voltage level for transmission-

1. Reduces volume of conductor material
2. Increases transmission efficiency
3. Decreases percentage line drop

Limitations of high transmission voltage

1. Increased cost of insulating the conductors
2. Increased cost of transformer and protection elements.

Bangladesh Power System

Present Structure of Power Sector

Generation Capacity : 32,332 MW *
Highest Generation : 17,200 MW (20 May, 2026)
Total Consumers : 49.7 Million
Transmission Line : 17,995 Ckt. km
Distribution Line : 6,57,266 km
Distribution Loss : 7.38% (June 2025)
Per Capita Generation : 661 KWh (FY 24-25)
Access to Electricity : 100%
Solar Home System (SHS): 6 million
*With Captive & RE

Consumption by sector (% of total)	
Residential	56.42% FY21
Industrial	28.40% FY21
Agriculture	2.43% FY21
Commercial and public sector	12.74% FY21

Source: <https://powercell.gov.bd/site/page/d730f98d-8912-47a2-8a35-382c4935eddc/Power-Sector-at-a-Glance>

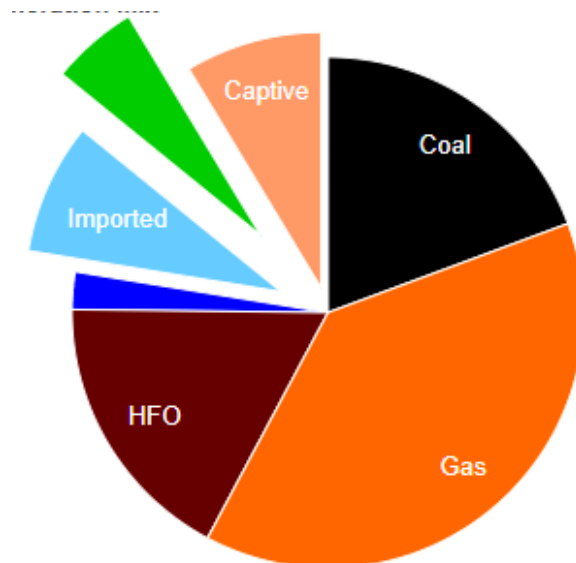
Transmission lines use "CKM" (circuit kilometers) to account for the number of circuits on a single physical route, while distribution lines use "km" (kilometers). Transmission lines, often spanning long distances and carrying high voltage, can have multiple circuits (2 or 3 phase, etc.) supported on the same towers. This means the total length of conductors is greater than the physical distance between the towers. Distribution lines, which serve local areas at lower voltages, generally have only one circuit per physical length.

A captive power plant (CPP), also known as autoproducer or embedded generation, is an electricity generation facility owned and operated by an industrial or commercial user for their own consumption.

Bangladesh Power System

Present Structure of Power Sector

Fuel/Resource	Installed Capacity	Share
Coal	6273 MW	19.34 %
Gas	12472 MW	38.45 %
HFO	5641 MW	17.39 %
HSD	768 MW	2.37 %
Imported	2696 MW	8.31 %
Renewable	1783.53 MW	5.5 %
Captive	2800 MW	8.63 %
Total	32434 MW	



Source: <https://ndre.sreda.gov.bd/index.php?id=7>

Bangladesh Power System



The **Payra 1320 MW thermal power plant** is a 1,320 megawatt coal-fired power station built in Kalapara Upazila of Patuakhali District in southern Bangladesh. It is a joint venture between the North-West Power Generation Company Bangladesh Limited (NWPGL) and China National Machinery Import and Export Corporation (CMC).



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1 基本数据	Basic Technical Data	
额定容量	Rated capacity	776.5 MVA
额定功率	Rated active power	660 MW
额定定子电压	Rated stator voltage	22 kV
额定定子电流	Rated stator current	20377 A
功率因数	Rated power factor	0.85 (lagging)
氢 压	Rated hydrogen pressure	0.45 MPa (g)
额定转速	Rated speed	3000 r/min

Bangladesh Power System



The biggest operating solar power plant in Bangladesh is the Teesta Solar Park in Sundarganj Upazila, Gaibandha District.

Capacity: 200 MW (AC)

Land Area: 650 acres of land in the Teesta river basin

Infrastructure: Consists of 550,000 solar panels

Investment: Approximately \$300 million

Operation Date: Commissioned and fully operating since August 2023

Grid Connection: Feeds power to the Rangpur grid via a 34 km transmission line

Bangladesh Power System

Useful Website Links for information related to Bangladesh Power System

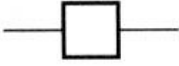
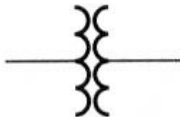
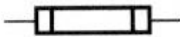
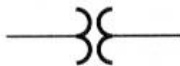
1. EMR: <https://emrd.gov.bd/>
2. Power Division: <https://powerdivision.gov.bd/>
3. Power Cell: <http://www.powercell.gov.bd/>
4. BPDB: <http://www.bpdb.gov.bd/>
5. PGCB: <http://pgcb.gov.bd/>
6. SREDA: <http://www.sreda.gov.bd/>
7. EPRC: <http://eprc.gov.bd/>

Representation of Power System

Symbols

TYPICAL COMPONENTS OF A POWER SYSTEM

- ❖ GENERATOR
- ❖ CIRCUIT BREAKER
- ❖ TRANSFORMER
- ❖ CURRENT TRANSFORMER
- ❖ POTENTIAL TRANSFORMER
- ❖ TRANSMISSION LINES
- ❖ DISTRIBUTION LINES

Rotating machine		Oil (or liquid) circuit breaker	
Two-winding power transformer		Air circuit breaker	
Three-winding power transformer		Delta connection	
Fuse		Wye connection, neutral ungrounded	
Current transformer		Wye connection, neutral grounded	
Potential transformer			

Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM

- ❑ INDICATED BY A SINGLE LINE AND STANDARD SYMBOLS
- ❑ INDICATES HOW THE TRANSMISSION LINES AND ASSOCIATED APPARATUS ARE CONNECTED TOGETHER
- ❑ PROVIDES SIGNIFICANT INFORMATION ABOUT THE SYSTEM IN CONCISE FORM

Since a balanced three-phase system is always solved as a single-phase circuit composed of one of the three lines and a neutral return, it is seldom necessary to show more than one phase and the neutral return when drawing a diagram of the circuit. Often the diagram is simplified further by omitting the completed circuit through the neutral and by indicating the component parts by standard symbols rather than by their equivalent circuits. Such a simplified diagram of an electric system is called a one-line diagram. It indicates by a single line and standard symbols the transmission lines and associated apparatus of an electric system.

Applications:

- Short circuit calculations
- Load flow studies
- Electrical safety procedures
- Efficient maintenance

Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM

Construction of single line diagram

1. As we know that the single line diagram is a simplified way for representing a 3 phase power system, so in case of representing each of the 3 phases with a separate terminal only one conductor is used to represent it.
2. The electrical elements like transformer, capacitor, circuit breaker, bus bars are represented by the standard schematic symbols.
3. It is to be noted that the elements portrait on the diagram does not represent actual physical size of the electrical component.

Single Phase representation of a Three Phase balanced system

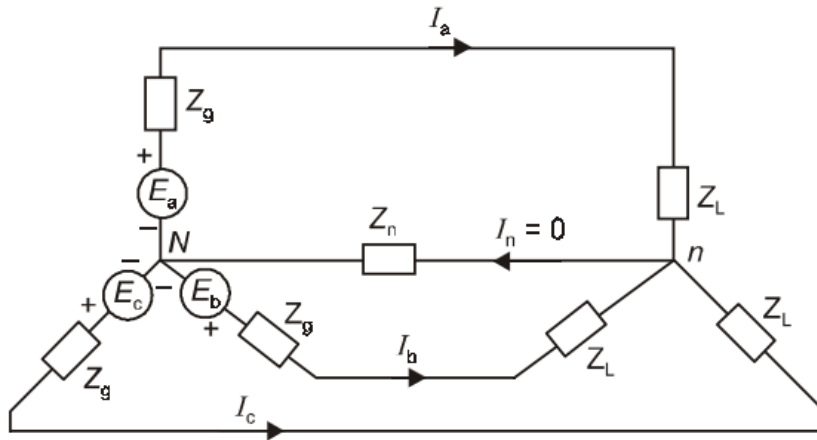


Fig. 5.1: Balanced three phase network.

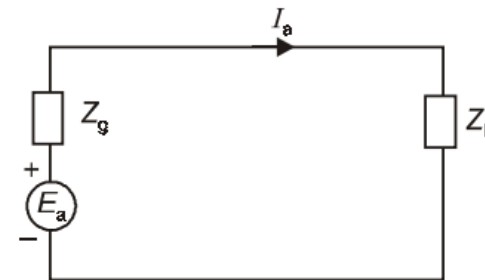
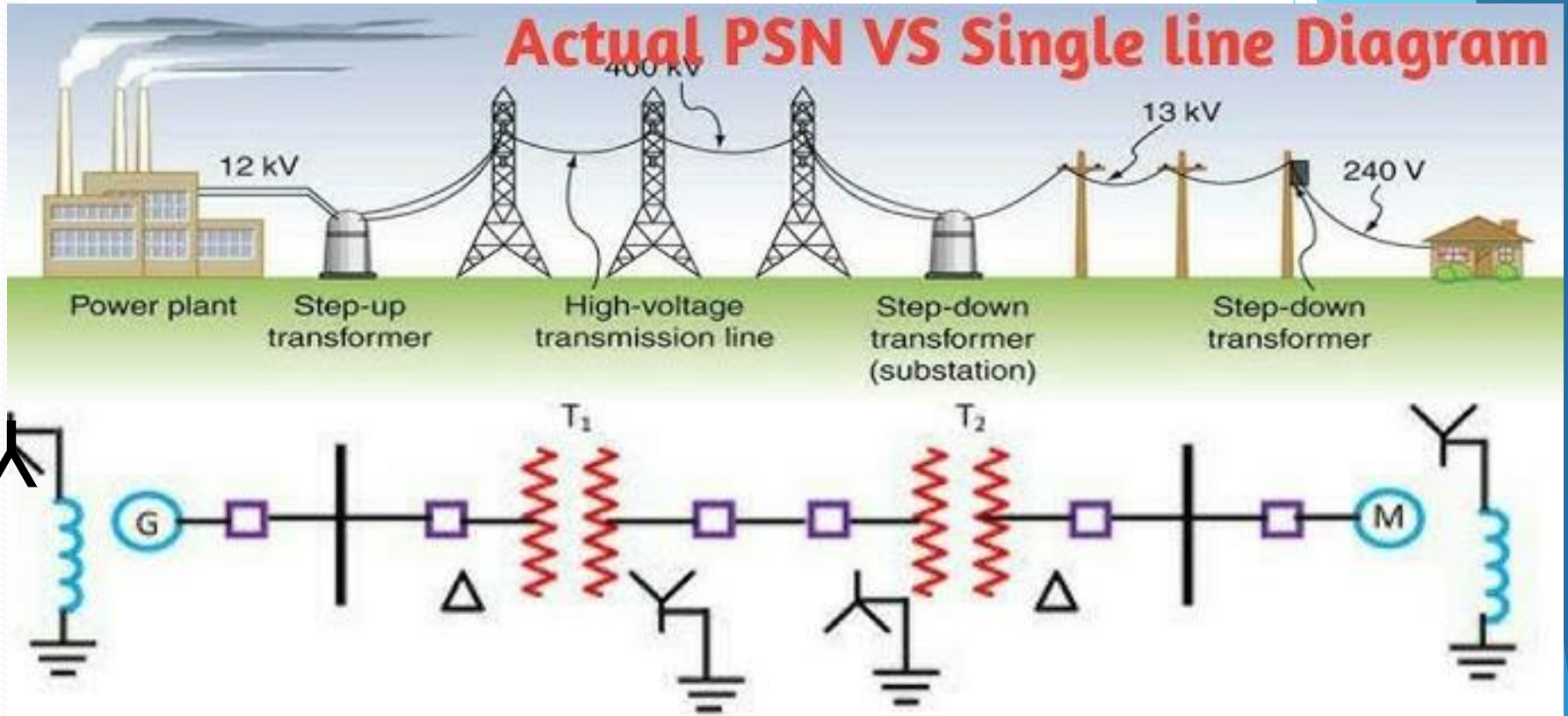


Fig. 5.2: Single-phase representation of a balanced three phase network of Fig. 5.1.

Representation of Power System

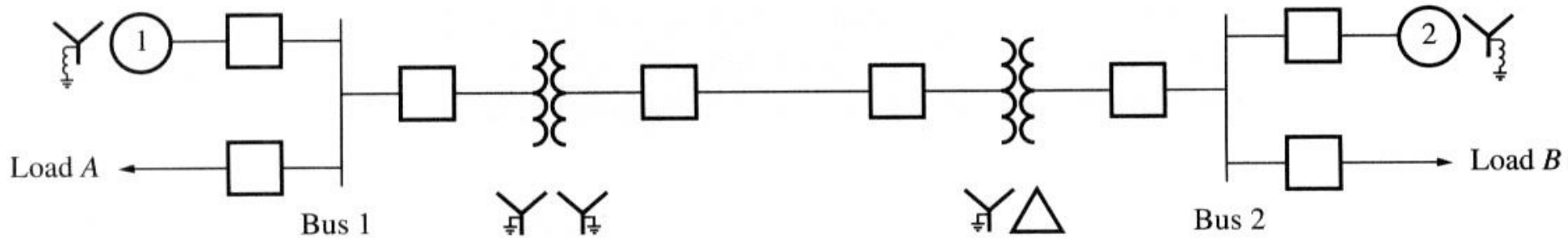
SINGLE LINE OR ONE LINE DIAGRAM

Actual PSN VS Single line Diagram



Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM



The figure shown above shows an example of power system containing the following components:

- Two synchronous machines (machines 1 and 2)
- Two loads (load A and load B)
- Two buses (Bus 1 and Bus 2)
- Two transformers
- Transmission line to connect buses together

Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM



#1 Generator—20,000 kva, 6.6 kv, $X'' = 0.655$ ohms

#2 Generator—10,000 kva, 6.6 kv, $X'' = 1.31$ ohms

#3 Generator—30,000 kva, 3.81 kv, $X'' = 0.1452$ ohms

T_1 and T_2 —each transformer in each 3-phase bank—10,000 kva, 3.81–38.1 kv,
 $X = 14.52$ ohms referred to the high-tension side

Reactance of the transmission line = 17.4 ohms

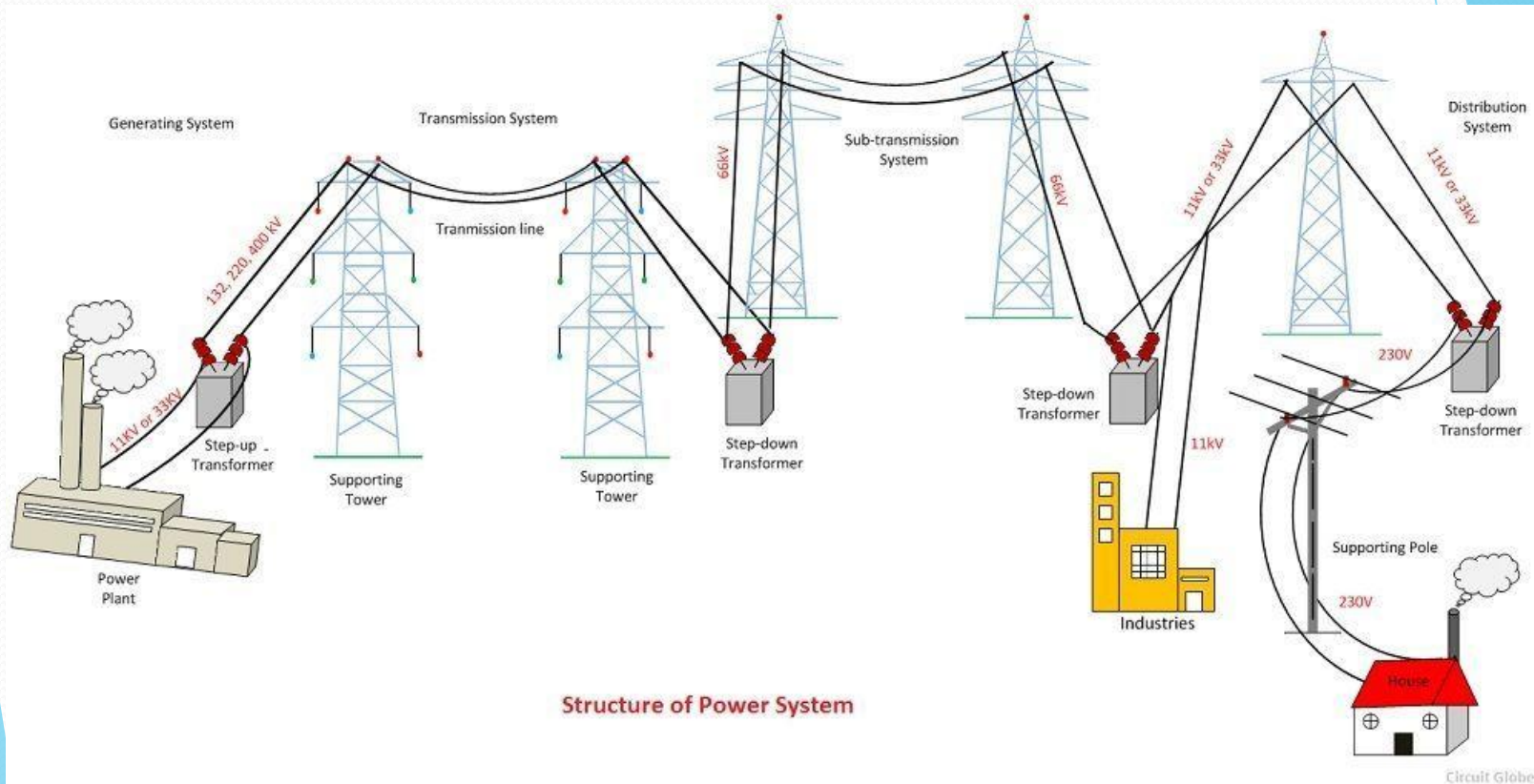
Load A = 15,000 kw, 6.6 kv, p.f. = 0.9 lag

Load B = 30,000 kw, 3.81 kv, p.f. = 0.9 lag

Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM

Exercise:





QUESTIONS AND DISCUSSIONS

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